



RESTORATIONESSENTIALS

Factory-Correct Restoration Reference

LS/LT Conversion Guide — 1960–1972 GM C10 Trucks

Model Year 2026

Recipient trucks	Chevrolet/GMC C10–C30 pickups (1960–72, all three generations)
Donor engines	Gen III/IV LS truck donors · crate LS3 · Gen V LT
Highlights	1960–62 fab flagged honestly; in-cab tank decision chapter; worksheets
Price	\$29.95 — PDF download at checkout

Every specification in this guide was checked against factory literature, marque references, period road tests, and documented survivor cars. Where sources disagree or the record is silent, this guide says so rather than guessing.

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Chapter 1 — The Decision: Donor, Crate, or Gen V LT (Read This First)

The 1960–72 GM truck is where the LS swap stops being one job and becomes three: the torsion-bar 1960–62, the coil-spring 1963–66, and the 1967–72 each accept the engine differently. Shops love these trucks because the bays are honest and the sheet metal is worth saving; builders bleed on them because a part that 'fits C10' fits one generation and mocks the other two. This chapter prices the whole job, per generation, before the first bolt turns.

- **Donor-first (the default, and this guide's main road):** a salvage truck 5.3 or 6.0 (LM7/LQ4/LQ9 family) with its harness, PCM, and accessories. The truck donor's tall intake and high alternator meet an earlier, lower hood line than the squarebody's, so drive choice is a real decision here (Chapter 3), not an inheritance.
- **Crate:** a new LS3 or controller-package crate. It buys engine certainty only; the tank, brakes, fuel, and wiring bills below are the same either way — and on these trucks the 'everything else' list is longer than the engine. Run Worksheet C before romanticizing either path.
- **Gen V LT:** the direct-injected L83/L86 truck engines. Real efficiency and torque; also high-pressure fuel hardware, VVT/AFM calibration behavior, and its own chapter (Chapter 4). An LT into a truck whose tank still lives behind the seat is two projects wearing a trench coat — plan both or pick the LS.

The line items first-timers miss (planning ranges, not promises — salvage and parts prices move): mounts (and on a 1960–62, fabrication the later trucks don't need); a transmission tunnel decision; the fuel system end to end, starting with the in-cab tank question; brakes — a four-drum, single-circuit 1960s truck is not a stopping system for 300 horsepower, and this guide treats that as a line item, not an upsell; cooling; the harness and PCM edits; exhaust by-year fitment; and a driveshaft measurement. If the honest total busts the budget, the endorsed alternative is finishing the truck with the drivetrain it has — a running truck at the cruise beats a cab off its mounts in the garage.



Note. If you only have \$X: spend it in this order — (1) stopping (front discs, dual-circuit master), (2) mounts that sit the engine honestly, (3) the fuel system done once at full pressure spec, (4) cooling, (5) the PCM edits, (6) cosmetics. Nobody at the cruise night asks what your fuel pressure is; everybody notices whether you arrived.

Chapter 2 — What You Start With: Three Generations of Baseline

Identify your generation first; then run the baseline against it. Pass/fail criteria, not vibes:

- **1960–62:** torsion-bar front suspension, its own frame with different crossmember and cab-mount arrangement. The later trucks' bolt-in parts do not bolt in here. Fabrication or swap-crossmember budget is a Chapter 1 line, decided now.
- **1963–66:** coil-spring front (the design that ran, with changes, through 1987). Adapter-plate support is real; frame horns and cab mounts still differ from 1967–72 — buy by year, not by nameplate.
- **1967–72:** the most-supported generation; still splits at 1971 for front discs and the 5-lug change, and at the tunnel (low-hump vs high-hump floors) for any big transmission.
- **Frame truth, all years:** trailing-arm rear (Chevrolet coils) vs leaf (most GMCs and 1-tons) changes the exhaust and driveshaft conversation; rust at the cab mounts and rear cab corners decides whether this is a swap or a restoration wearing one.
- **Brakes — go/no-go:** four drums and, on pre-1967 trucks, a single-circuit master cylinder. With swap power, front discs and a dual-circuit master move from 'someday' to baseline. No road speed without them (Chapter 13 enforces it).
- **Lug pattern honesty:** 6-lug through 1970, 5-lug (5×5) from 1971 on 2WD half-tons. A disc-brake conversion may change your wheels — worksheet item, not a surprise at the tire shop.
- **The donor audit** (family discipline): compression or a running demonstration, oil condition, and a signed accessory inventory — alternator, starter, MAF, DBW pedal, PCM, harness completeness. 'Complete pullout' is a sales phrase; the checklist is the contract.

Chapter 3 — Fitment: 1960–62 Is Not 1967–72

Mounts. On 1963–72 coil-spring frames, adapter plates on the stock stands put the LS where the small-block lived and keep the transmission in its stock position — tunnel, shaft length, and linkage stay factory problems you already solved. On the 1960–62 torsion-bar frame the mounting geometry is its own; published builds on these years start with mount and crossmember fabrication, and this guide says so plainly: budget the fab or the aftermarket crossmember, and don't let a parts counter sell you 1967–72 bolt-ins for a 1961.

Height is the early trucks' real conversation. The LS truck intake and truck accessory drive are tall; the 1960s hood line is not. Mock up with the hood closed and measured (Worksheet B): car-style drives and lower intakes are the fix when the truck stack kisses the hood — a parts decision made at mock-up, not after paint.

- **Oil pan and steering:** pans get verified at mock-up against the crossmember and the linkage at full lock, both directions — the early steering geometry differs by generation, which is why this is a measurement and not a table.
- **Tunnels:** 1967–72 low-hump floors were drawn around smaller cases; the high-hump (large) floor is the big transmission's friend. 4L80E, 6L80, and big manuals get measured against YOUR hump. 1960–66 tunnels predate the overdrive era entirely — assume the conversation, be pleased if it clears.



- **Manual-truck linkage:** clutch linkage and the steering column share the driver's-side real estate with the exhaust on these trucks; keep them in the mock-up picture (Chapter 8).
- **What does NOT need cutting, 1963–72:** firewall and inner fenders for a standard LS placement. If the torch comes out on a coil-spring truck, the plan skipped this chapter.

Chapter 4 — The Gen V LT Chapter: Direct Injection Changes the Job

The Gen V truck engines (L83 5.3, L86 6.2) sit on the same mounts and answer to the same transmission logic as the LS. Downstream of the fuel door, everything changes: a cam-driven high-pressure pump fed by a serious in-tank supply; vacuum management where your brake booster expects manifold vacuum; VVT and AFM in the calibration; and a harness-year matching exercise fussier than the LS. On a truck whose fuel system starts as a mechanical pump and an in-cab tank, the LT multiplies Chapter 6 — budget the lift-pump supply to hold the DI pump's inlet spec at full load, and let Chapter 1's decision be an honest one. Steam vents still get plumbed; DI did not retire physics.

Chapter 5 — Wiring and Controls

Harness strategy — the family's three roads: reworked donor harness (cheapest, most hours), standalone swap harness (the middle road), or a controller package with a crate engine. Whatever the road: the anti-theft (VATS/passkey) edit happens before first start, or the truck cranks forever in silence; fan outputs get programmed — truck PCMs do not drive electric fans by default; rear-O2 and emissions handling follows your jurisdiction (stated, never guaranteed).

- **Drive-by-wire pedal in a 1960s cab:** the floor and pedal geometry are the donor's, not yours — mock the mount, verify full travel to the stop, and drive geometry before final weld/rivets. The PCM reads angle, not effort; a bad mount is a forever drivability complaint.
- **Gauges:** period clusters take sender adaptation (oil/temperature), tach scaling by cylinder-count output, and the honest problem — a mechanical speedometer cable meeting a vehicle-speed sensor: electronic speedo, cable-drive conversion box, or GPS unit, chosen by budget.
- **Grounds:** engine-to-frame, engine-to-body, new straps on clean metal. 60-year-old cab grounds are the swap world's favorite unexplained electrical gremlin; remove the mystery before it moves in.

Chapter 6 — The Fuel System: The Tank Behind the Seat Has to Go EFI

The LS wants a regulated 58 psi at the rail, EFI-rated line end to end, and filtration (coarse before the pump, fine after). What a 1960–72 truck brings: a tank behind the seat, a mechanical pump on the old engine, and venting designed for a carburetor's modest drama. The tank decision is this chapter, and it is a decision — two honest roads:

- **Keep the in-cab tank, go EFI inside it:** in-tank modules exist for the factory tank. Preserves the truck's originality; the honest trade is fuel inside the cab at 58 psi — every fitting and line gets the inspection discipline in the safety note below, forever, not once.
- **Relocate to a rear-frame EFI tank:** the popular route. A baffled tank with an in-tank pump goes between the frame rails (under-bed); the in-cab tank retires, the vent routes outside, and the filler moves to the bed or fender. Costs the spare-tire real estate or bed work depending on kit — planned, measured, and photographed.
- **Either way:** Corvette-style filter/regulator near the tank (~58 psi, returnless rail) or a rail-regulated return; rollover vent routed safely out; no carb-era rubber anywhere in the pressure path.



Note. Fire discipline, plainly: 58 psi finds weak hose like water finds a roof leak. EFI-rated hose and proper clamps only — on an in-cab tank this is not a parts-preference, it is the whole safety case. Prime at pressure, leak-check with eyes and nose before first crank, re-check after the first heat cycle and at every 500-mile list.

Chapter 7 — Cooling and the Steam Lines

The LS non-negotiable stands: steam vents at the heads plumbed to the pressurized side, or trapped air makes steam pockets and a calm-looking gauge while the heads suffer. The early trucks' radiators were sized for their original engines and their narrower cores make the radiator-and-fan package a decision, not an inheritance: core condition and capacity first, fan second (mechanical clutch fans are honest and simple; electrics demand a shroud that actually fits — measured), capacity last. Diagnosis order for any swap that runs hot: steam vents, airflow, capacity, tune.

Chapter 8 — Exhaust: Headers by Year, Manifolds by Honesty

Header fitment on these trucks is year-sensitive — 1960–66 and 1967–72 are different purchases, and on 1960–62 the torsion-bar frame writes its own rules again. Swap-header lines exist for the coil-spring trucks; manual-clutch linkage and the steering column compete with the driver's header for the same inches, so the mock-up includes them. And the family's standing honesty rule: the donor's manifolds lose some power and clear everything — on a working classic truck that is frequently the correct answer, and this guide will not pretend otherwise. Ground clearance on a lowered truck interviews the collectors first; mid-lengths exist for a reason.

Chapter 9 — ECU and Tuning Decisions

A stock calibration carrying the swap edits (anti-theft out, fan outputs on, gear/tire math corrected, rear-O2 handling per your jurisdiction) runs a stock donor engine the way the donor truck ran. A mail-order tune covers the edit list and mild combinations; it cannot fix a cam it has never seen. A dyno tune earns its money on cams, forced induction, and combinations — and is sold as habit on stock swaps. Decide by what changed.

Chapter 10 — The Gotchas

Each gotcha in the family's fixed format: *What happens / Why it happens / How to see it coming / The fix, in order of cost*. Categories the spec requires are all here; a category clear on this body says so in one line.

- **'Fits C10' parts on the wrong generation.** *What:* the bolt-in kit doesn't. *Why:* 1960–62, 1963–66, and 1967–72 frames differ in mounts, horns, and crossmembers. *See it:* Chapter 2's generation ID before ordering anything. *Fix, in cost order:* buy by year; on 1960–62, budget fabrication rather than discovering it.
- **The tall truck intake meets the low hood.** *What:* the truck intake/drive stack kisses the hood. *Why:* the donor's bay is taller than yours. *See it:* mock-up with the hood closed, measured (Worksheet B). *Fix:* car-style drive/intake components; mount-position tuning — before paint, not after.
- **The in-cab tank, unbaffled and unpressurized.** *What:* starvation on corners and a venting system from the carburetor era. *Why:* sixty years of design intent. *See it:* Chapter 6 — every one of these trucks. *Fix:* in-cab EFI module (with the safety discipline) or the rear-frame relocation tank.
- **Big overdrive, small tunnel.** *What:* the 4L80E/6L80 meets the low-hump floor. *Why:* the tunnels predate the big cases. *See it:* Chapter 3's hump measurement. *Fix:* transmission choice first, crossmember position second, floor work last — permanent is last.



- **Header vs steering and clutch linkage.** *What:* the driver's header shares inches with the box, column, and clutch linkage. *Why:* 1960s packaging. *See it:* mock-up with all three in place. *Fix:* year-correct swap headers, or the manifolds that clear everything (Chapter 8's honest trade).
- **The six-lug/five-lug cascade.** *What:* the disc conversion changes the wheel pattern and the old wheels retire mid-project. *Why:* 1971 changed both at once. *See it:* Chapter 2's lug inventory. *Fix:* decide the brake/wheel package as one Worksheet C line.
- **Throttle-pedal geometry (DBW).** *What:* a donor pedal at 1960s-cab angles drives lazy or touchy forever. *Why:* the PCM reads angle, not effort. *See it:* mocked mount, full travel verified. *Fix:* reposition; no tune cures geometry.
- **Donor accessory completeness.** *What:* the 'complete pullout' lacks the pedal, MAF, or PCM. *Why:* 'complete' is a sales word. *See it:* Chapter 2's signed checklist. *Fix:* inside the negotiation, never after it.
- **Steam pockets.** Chapter 7 — graded here: applies to every LS/LT swap; vent the heads.
- **Driveshaft and carrier bearing (long-beds).** *What:* vibration after the transmission change. *Why:* two-piece shafts and their angles were set for the old driveline. *See it:* Worksheet A at final ride height. *Fix:* angles measured and corrected; shaft built to your measurement, never a forum number.
- **Checked — clear on this body:** shock towers (there are none to notch); firewall clearance for a standard LS placement on 1963–72 (1960–62 verifies at mock-up like everything else on that frame).

Chapter 11 — Tips and Tricks

- **Order parts by year and generation, never by nameplate.** 'C10' spans three frames; the box doesn't know which one is in your garage. You do (Chapter 2).
- **Plan on three engine installations:** mock-up, clearance-check, final. The builders who do it once planned on three and got lucky twice.
- **Do the brake conversion while the front clip is apart** — the same Saturday, the same momentum, not a second teardown.
- **Bag and tag the donor teardown;** the connector you cut 'because it looked unused' is the one the PCM wants.
- **Photograph every routing decision;** heat and chafe are patient, and the photo is the map at the first repair.
- **Chase threads before torque values** — 60-year-old fasteners lie to torque wrenches.

Chapter 12 — Special Tools

Must-have: a hoist with an engine leveler (the engine goes in at an angle over a tall core support; the leveler is the tool, the hoist is its helper), rated stands, a fuel-pressure gauge that reads over 60 psi (Chapter 6's verification is not optional), a multimeter, thread chasers. **Rent or borrow:** the transmission jack, the balancer puller/installer, brake-line flare tools for the disc conversion, a ball-joint press if spindles are changing. **Nice:** a lift — it converts Chapter 13's under-truck checks from an afternoon to an hour. No tool is listed without the chapter that uses it.

Chapter 13 — First Start and Shakedown

- **Before cranking:** fuel system primed and leak-checked at full pressure (double-length, slow inspection if the tank stayed in the cab — Chapter 6's discipline); vents verified; oil primed; PCM security status confirmed; extinguisher in reach.
- **Brakes before road speed, full stop:** disc/dual-circuit conversion bled and verified at walking pace, then neighborhood pace. The first drive is a test, not a commute.



- **First start:** one person at the key, one at the bay. Short cranks; oil pressure is the only gauge today.
- **First heat cycle:** fan trigger against a real temperature; steam vents re-checked; every fitting in the fuel path re-inspected warm.
- **50 miles:** leaks, belts, chafe, driveline report (Worksheet A if it vibrates). **500 miles:** re-torque list, fluids, alignment after springs settle — and Chapter 9's honest conversation about what the tune still owes you.

Note. Safety, plainly: a single-circuit brake master on a pre-1967 truck is named here as replace-before-road-speed, not 'period charm.' Steering fasteners disturbed in the swap torque to the factory literature on inspected metal — never 'snug by feel' on the parts that aim the truck.

Appendix: Worksheets (Fill These In On YOUR Car)

Worksheet A — Driveline. Two-piece or one-piece shaft: _____. Carrier bearing present (Y/N): _____. Ride height, four corners as driven: _____. Output angle: _____. Pinion angle: _____. Working angles: _____. Shaft length, center-to-center at height: _____.

Worksheet B — Tunnel and clearance. Generation (1960–62 / 1963–66 / 1967–72): _____. Tunnel hump (measured): _____. Transmission chosen: _____. Bellhousing-to-floor clearance: _____. Hood-closed clearance over the intake/drive: _____. Pan-to-linkage clearance at full lock L/R: ____ / _____. Header-to-steering/clutch clearance: _____.

Worksheet C — Parts-total ledger. Engine/donor: _____. Mounts (and 1960–62 fab): _____. Brakes (discs, master, lines): _____. Fuel system (tank decision, pump, lines, regulator): _____. Cooling: _____. Harness/PCM/tune: _____. Exhaust: _____. Transmission and driveshaft: _____. Wheels/lugs if the brake package changes them: _____. Total, honest: _____. Compare against the crate quote and against finishing the truck as-is — that comparison IS Chapter 1's decision.

Recommended for this restoration

Shop exact-fit supplies on Amazon

- LS swap C10 1967-1972 motor mount kit
- 1960-1972 C10 rear frame EFI fuel tank conversion
- 1967-1972 C10 front disc brake conversion kit
- LS swap standalone wiring harness

As an Amazon Associate, RestorationEssentials earns from qualifying purchases made through these links.

Sources Consulted

Every specification in this guide was checked against published references before printing. Where sources disagree, this guide says so in the text rather than picking a number silently. The references below are the backbone of the technical chapters:



- **GM C/K platform documentation.** 1960–72 truck platform history: the 1960–62 torsion-bar frame, the 1963 coil-spring front design and its running changes (1971 front discs, spindle/ball-joint revisions, the 5-lug change), trailing-arm vs leaf rear applications
- **Frame-generation comparisons, 1960–66 vs 1967–72.** published frame and cab-mount comparisons and documented 1960–62 swap builds (mount and crossmember fabrication on the torsion-bar trucks) from the 67-72chevytrucks reference community
- **Manufacturer fitment documentation.** adapter-plate and swap-header fitment by year (the year-specific notes for 1960–66 vs 1967–72) as published by the swap-parts makers
- **GM LS-family production data.** engine RPO applications and factory ratings for Gen III, Gen IV, and Gen V engines as published in GM engine-family references
- **EFI system manufacturers' installation literature.** fuel-pressure, filtration, and in-tank module requirements as published for LS conversions

A note on numbers. Factory literature, period road tests, and marque historians do not always agree — production totals, horsepower ratings, and paint codes all have documented conflicts. This guide uses the best-supported figure in each case and flags the conflict where one exists, so you can judge for yourself.

Emissions legality varies by jurisdiction — even pre-emissions-era trucks live under state rules where states have them: explained, never guaranteed. Structural rust and frame repair belong to the restoration guides and to shops; forced induction beyond the factory-supercharged engines is outside this guide's scope.



Thank You

Thank you for choosing this RestorationEssentials guide. Whether this is your first LS/LT Conversion — GM C10 1960–1972 or your fiftieth, you picked one of the great ones, and the fact that you are doing your homework before you turn a wrench puts you ahead of most of the field.

These cars survived because people like you refused to let them disappear. Every correct fastener, every properly decoded tag, every hour spent getting the details right keeps the history honest for the next owner and the next generation of enthusiasts.

Modern engine, honest old truck — planned by generation, not by nameplate. Measure on your truck, write it in the worksheets, and finish it: a finished swap beats a perfect plan every Saturday night.

If this guide helped your build, we would love to hear about it. Good luck with the restoration, and enjoy every mile.

— The RestorationEssentials Team